Yang Liu

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PROFESSIONAL EXPERIENCE

University of Cambridge - Department of Computer Science and Technology, *United Kingdom*
Senior Research Associate — *Feb. 2024 – Present*
Advisor: Prof. Cecilia Mascolo

Affiliated Lecturer — *Oct. 2022 – Present*

University of Cambridge - Department of Computer Science and Technology, *United Kingdom*
Research Associate — *Feb. 2022 – Jan. 2024*
Advisor: Prof. Cecilia Mascolo

University of Cambridge - Newnham College, *United Kingdom*
Postdoctoral Affiliate
The oldest college run by women, for women — *Oct. 2022 – Sep. 2024*

City University of Hong Kong - Department of Computer Science, *Hong Kong SAR*
Postdoctoral Researcher — *Oct. 2020 – Dec. 2021*
Advisor: Dr. Zhenjiang Li

EDUCATION

City University of Hong Kong, *Hong Kong SAR*
Ph.D. in Computer Science — *Sep. 2016 – Oct. 2020*
Advisor: Dr. Zhenjiang Li

Xi'an Jiaotong University, *Shaanxi, China*
B.Eng. in Software Engineering — *Sep. 2012 – Jun. 2016*

RESEARCH INTERESTS

My research focuses on developing intelligent mobile and wearable technologies for human-centered sensing and computing. It explores and expands the perception capabilities of existing wearable devices to deepen and broaden innovative ways for understanding human behavior, monitoring human wellness, and facilitating interactions between humans and machines or ambient environments. By gathering insights into individual behavior, health, and interactions with mobile and wearable systems, my research advances cutting-edge areas such as mobile health, Human-Computer and Human-Ambient Interactions, the Internet of Things (IoT) and cybersecurity. The methodologies of my research encompass sensory data analysis and modeling, signal processing, machine learning, system design and optimization, as well as the development of prototypes and testbeds for evaluation.

HONORS & AWARDS

- N2Women: Rising Stars in Computer Networking and Communications [*one of ten global awardees*] — *2024*
- Best Paper Award – ACM EarComp — *2023*
- Outstanding Doctoral Thesis Award – ACM SIGBED China [*one of three awardees in China*] — *2021*
- Institutional Research Tuition Scholarship – City University of Hong Kong (CityU) — *2019*
- Outstanding Academic Performance Award – CityU — *2018, 2019*
- Student Travel Grant – ACM MobiSys — *2019*
- Best-in-Session Presentation Award – IEEE INFOCOM — *2018*
- Student Travel Grant – IEEE INFOCOM — *2018*
- Outstanding Poster Award – CityU — *2017*

TEACHING

- Mobile, Wearable Systems and Machine Learning, University of Cambridge
  Lecturer — *Michaelmas 2024/2025, Michaelmas 2023/2024*
- Computer Networking, University of Cambridge
  Group Supervisor — *Lent 2024/2025, Lent 2023/2024, Lent 2022/2023*
- Computer Programming, City University of Hong Kong
  Lab Tutor — *Sem B 2019/2020, Sem B 2018/2019*
  Teaching Assistant — *Sem A 2017/2018, Sem A 2016/2017*

- Introduction to Computer Studies, City University of Hong Kong
  Teaching Assistant *Sem A 2019/2020*
- Computer Networks, City University of Hong Kong
  Tutorial Tutor *Sem A 2018/2019*
- Cloud Computing, City University of Hong Kong
  Teaching Assistant *Sem B 2017/2018*
- The Art and Science of Data, City University of Hong Kong
  Teaching Assistant *Sem B 2016/2017*

MENTORSHIP

- Yanbin Gong (The Hong Kong University of Science and Technology) – Early Detection of AECOPD Using a Smartphone
  (*under preparation*) *Nov. 2024 - Present*
- Jiani Cao (City University of Hong Kong) – A Wearable EEG Foundation Model for Motor Imagery
  (*under preparation*) *June. 2024 - Present*
- Jake Stuchbury-Wass (University of Cambridge) – WalkEar: Holistic Gait Monitoring using Earables
  (under review) *Sep. 2023 - Sep. 2024*
- Kayla-Jade Butkow (University of Cambridge) – Measuring Cardiac Stroke Volume Through In-ear Audio Sensing
  (under review) *Sep. 2023 - Sep. 2024*
- Jordan Waters (University of Cambridge) – Deep-Learning Based Segmentation of In-Ear Cardiac Sounds
  (under review) *Oct. 2023 - July 2024*
- Jiani Cao (City University of Hong Kong) – Finger Tracking Using Wrist-Worn EMG Sensors
  IEEE Transactions on Mobile Computing 2024 *Feb. 2023 - May 2024*
- Tamisa Ketmalasiri (University of Cambridge) – IMChew: Chewing Analysis using Earphone Inertial Measurement Units
  ACM BodySys 2024 *Oct. 2023 - May 2024*
- Jiani Cao (City University of Hong Kong) – ASGaze: Gaze Tracking on Any Surface with Your Phone
  ACM SenSys 2022, ACM TURC 2023, IEEE Transactions on Mobile Computing 2024 *April 2022 - May 2024*
- Jacob Brown (University of Cambridge) – Yawning Detection using Earphone Inertial Measurement Units
  ACM SmartWear 2023 *Oct. 2022 - July 2023*
- Jiao Li (City University of Hong Kong) – EarPass: Continuous User Authentication with In-ear PPG
  ACM EarComp 2023 [Best Paper Award] *Oct. 2022 - July 2023*
- Jiao Li (City University of Hong Kong) – GASLA: Enhancing the Applicability of Sign Language Translation
  IEEE INFOCOM 2022, IEEE Transactions on Mobile Computing 2024 *Oct. 2021 - July 2023*

PROFESSIONAL SERVICE

Organizing Committee
- Computer Science Ethics Committee, University of Cambridge *2024, 2025*
- Chairs, IEEE SWC *2024*
- Publicity & Social Media Chairs, ACM UbiComp/ISWC *2024*
- Publicity Co-chairs, ACM MobiCom *2022*
- Web Co-Chairs, IEEE ICPADS *2021*

Technical Program Committee
- ACM SenSys *2025*
- ACM MobiSys *2025*
- IEEE ICDCS *2024, 2023*
- ACM MobiSys (Artifact Evaluation) *2024, 2023*
- ACM MobiCom (Artifact Evaluation) *2024, 2023*
- IEEE ICPADS *2024, 2022*
- ACM ISWC *2024*
- ACM IPSN (Poster) *2024*
- IEEE TrustCom *2024*
- IEEE UIC *2024*
- ACM Workshops: EarComp, SmartWear, BodySys, HumanSys, EIFCom, MobiCom4AgeTech *2022-2024*
- ACM SenSys (Shadow Program) *2022*
- IEEE ICDCS (Ph.D. Symposium) *2022*
- EAI Qshine *2020*

Reviewer
- Nature Communications
- ACM CHI
- ACM IMWUT journal (UbiComp)
- ACM Transactions on Sensor Networks
- ACM Transactions on Internet of Things (TIOT)
- IEEE Transactions on Mobile Computing
- IEEE Internet of Things Journal (IoT-J)

- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Network Science and Engineering
- IEEE ICPADS 2021
- Journal of Ambient Intelligence and Humanized Computing
- Sensors Journal
- Applied Science Journal
- pro² network+ funding

PRESENTATIONS & TALKS

- Less is More: Innovative Wearable Sensing for Understanding Human Behavior and Monitoring Wellbeing
  North Carolina State University, October 2024, Virtual Event.
- Respiratory Rate Monitoring from Earables
  MobiUK, July 2024, Southampton, UK.
- IMChew: Chewing Analysis using Earphone Inertial Measurement Units
  MobiUK, July 2024, Southampton, UK.
- Yawning Detection using Earphone Inertial Measurement Units
  ACM SmartWear, October 2023, Madrid, Spain.
- EaRR: Audio-Based Robust Respiratory Rate Monitoring from Earables
  Mobile Systems Lab, July 2023, Cambridge, UK.
- Human Behavior Sensing and Understanding with Sparse Wearable Sensors
  City University of Hong Kong, August 2020, Hong Kong, China.
- WR-Hand: Wearable Armband can Track User's Hand
  ACM UbiComp, September 2021, Virtual Event.
- Real-time Arm Skeleton Tracking and Gesture Inference Tolerant to Missing Wearable Sensors
  ACM MobiSys, June 2019, Seoul, South Korea.
- cDeepArch: A Compact Deep Neural Network Architecture for Mobile Sensing
  IEEE SECON, June 2018, Hong Kong, China.
- aLeak: Privacy Leakage through Context-Free Wearable Side-Channel
  IEEE INFOCOM, April 2018, Honolulu, HI, USA.

LEADED RESEARCH PROJECTS

Motion-resilient Breathing Volume Monitoring from Earables (under review) *August. 2023 – Sep. 2024*
Breathing volume is a crucial vital sign, but current methods for its estimation either require invasive equipment in clinical settings or fail to deliver reliable results consistently in daily life. Additionally, the estimation of breathing volume during exercise remains largely unexplored, despite its significant implications for fitness monitoring. In this work, we propose using the in-ear microphone of earphones to estimate breathing volume in both stationary and exercise scenarios, enabling portable, non-invasive, and continuous monitoring. Concretely, we innovatively propose a series of deep learning and signal processing techniques to extract breathing-related features for breathing volume estimation, which leverage the correlation between cardiovascular and respiratory systems of human body for heart-modulated breathing features extraction and uni-modal knowledge distillation for high-quality breathing sound features extraction.

Robust Respiratory Rate Monitoring with In-Ear Microphones (under review) *June. 2022 – Dec. 2023*
Since respiratory rate (RR) is a key marker of physical and mental health, continuous monitoring of respiratory rate can enable better understanding of health and wellbeing. However, continuously monitoring respiratory rate is still an open research problem that has yet to be solved. In this project, we are developing the first work that makes RR monitoring using earphones works under different scenarios of daily life. We make key findings that the body sounds captured via the inward-facing microphone on earphone are exactly in the right place to have different coupling relations with RR under different scenarios, which could allow us to enable reliable RR monitoring in much more generalised scenaris compared with the state-of-art works. A series of novel signal processing techniques have been proposed to realise this system. *A patent application has been filed for this work, and it is being commercialized with Cambridge Enterprise.*

Hand Pose Tracking with a Commercial Wearable Armband *Sep. 2019 – May. 2021*
We have enabled a wearable-based system that can track the 3D hand pose (14 hand skeleton points) over time with a commercial armband worn on user's forearm. Compared with state-of-arts using cameras or attaching many sensors on user's body, our solution is user-friendly, tolerant to darkness and occlusion, portable and lightweight to execute on user's mobile phone to directly support upper-layer applications, like neural-aided HCI, hand-motion rehabilitation and VR. We have addressed a challenge using a visualized DNN that the armband collects mixed Electromyography (EMG) signals from multiple forearm muscles, while prior bio-medical models are built on the isolated EMG signals from different muscles. Then, we placed the constructed hand pose in a global coordinate system by fusing the IMUs data, provided a general plug-and-play version for new users through combining with an adversarial network, and compensated for the position difference in how users wear their armbands through signal analysis. A demo video can be found **here**.

Real-time Arm Tracking and Gesture Inference via Sparse Wearable Sensors *May. 2018 – Sep. 2019*
A wearable system was proposed to achieve real-time 3D arm skeleton tracking with a smart watch only and reliable gesture inference tolerant to missing wearable sensors. This system can be directly deployed with users who have a smartwatch

and mobile phone, which can be widely adopted and benefit many useful applications, like healthcare for Parkinson and Alzheimer, HCI for smart home/car and VR. We have addressed two challenges. First, the skeleton of each arm is determined by the locations of the elbow and wrist, whereas a smart watch only senses a single point from the wrist. The potential solution space is huge, which challenges the accurate and real-time arm skeleton tracking. Here, we implemented arm tracking using the kinematic theory of human arm and the Hidden Markov Model algorithm, and then we proposed two acceleration algorithms to enable real-time tracking. Second, some wearable devices may be missing. Yet the learning tools for gesture inference typically have static network structures, which requires nontrivial network adaptation to match the input's varying availability and ensure reliable gesture inference. We here proposed an attention-based mechanism to achieve dynamic network adjustment. A demo video and *open-source code* can be found **here**.

Privacy Leakage through Context-Free Wearable Side-Channel *May. 2017 – May. 2018*
We revisited a crucial privacy problem: could the sensitive information, like passwords, frequently typed on mobile devices be inferred through the motion sensors of wearable device on user's wrist, e.g., smart watch? Existing works have achieved the initial success under certain context-aware conditions, such as 1) the horizontal keypad plane, 2) the known keyboard size, 3) and/or the last keystroke on a fixed "enter" button. Taking one step further, we unveiled and fully demonstrated the further risks of typing privacy leakage in much more generalized context-free scenarios without any above context-aware conditions. A series of novel IMU analysis algorithms is proposed to enable the design.

Object Size Estimation with Wearable Sensing and Crowdsourcing *Sep. 2016 – May. 2017*
We proposed a mobile system, which turned our wearable or mobile device into a ruler. It can estimate the size of objects that could be large in size and not directly touchable by the user. Such a design can enable a rich set of applications that count on the size information of surrounding environments/objects. We proposed several sensing techniques to purely utilize the motion sensors on the device for object size measure and also integrated with the crowdsourcing feature for both performance improvement and result sharing.

PUBLICATIONS

Name* is the student I mentored for those publications.

[1] **Yang Liu**, Qiang Yang, Kayla-Jade Butkow, Jake Stuchbury-Wass, Dong Ma, Cecilia Mascolo. "EarMeter: Continuous Respiration Volume Monitoring with Earable Audio", (under review).

[2] **Yang Liu**, Kayla-Jade Butkow, Jake Stuchbury-Wass, Adam Pullin, Dong Ma, Cecilia Mascolo. "RespEar: Earable-Based Robust Respiratory Rate Monitoring", Arxiv, 2024, (under review). [Under commercialization efforts with Cambridge Enterprise]

[3] Kayla-Jade Butkow*, Navazh Jalaludeen, **Yang Liu**, Jake Stuchbury-Wass, Qiang Yang, Mathias Ciliberto, Dong Ma, Joseph Cheriyan, Cecilia Mascolo. "Measuring Cardiac Stroke Volume Through In-ear Audio Sensing", (under review).

[4] Qiang Yang, **Yang Liu**, Jake Stuchbury-Wass, Kayla-Jade Butkow, Dong Ma, Cecilia Mascolo. "SmarTeeth: Augmenting Manual Toothbrushing with In-ear Microphones", (under review).

[5] Jake Stuchbury-Wass*, **Yang Liu**, Kayla-Jade Butkow, Josh Carter, Qiang Yang, Ezio Preatoni, Dong Ma, Cecilia Mascolo. "WalkEar: Holistic Gait Monitoring using Earables", (under review).

[6] Jordan Water*, Jake Stuchbury-Wass, **Yang Liu**, Kayla-Jade Butkow, Cecilia Mascolo. "Deep-Learning Based Segmentation of In-Ear Cardiac Sounds", (under review).

[7] Yuwei Zhang, Tong Xia, Jing Han, Yu Yvonne Wu, Georgios Rizos, **Yang Liu**, Mohammed Mosuily, Jagmohan Chauhan, Cecilia Mascolo. "Deep-Learning Based Segmentation of In-Ear Cardiac Sounds", NeurIPS, 2024.

[8] Qiang Yang, **Yang Liu**, Jake Stuchbury-Wass, Kayla-Jade Butkow, Dong Ma, Cecilia Mascolo. "BrushBuds: Toothbrushing Tracking Using Earphone IMUs", EarComp, 2024.

[9] Jiani Cao*, Jiesong Chen, Chengdong Lin, **Yang Liu**, Kun Wang, Zhenjiang Li. "Practical Gaze Tracking on Any Surface with Your Phone", IEEE Transactions on Mobile Computing, 2024.

[10] Jiani Cao*, **Yang Liu**, Lixiang Han, Zhenjiang Li. "Finger Tracking Using Wrist-Worn EMG Sensors", IEEE Transactions on Mobile Computing, 2024.

[11] Changshou Hu, Thivya Kandappu, **Yang Liu**, Cecilia Mascolo, Dong Ma. "BreathPro: Monitoring Breathing Mode during Running with Earables", ACM IMWUT journal (UbiComp) 2024. [Media Exposure: "SMU City Perspectives"]

[12] Changshuo Hu, Thivya Kandappu, Jake Stuchbury-Wass, **Yang Liu**, Anthony Tang, Cecilia Mascolo, Dong Ma. "Detecting Foot Strikes during Running with Earbuds", BodySys, 2024.

[13] Tamisa Ketmalasiri*, Yu Wu, Kayla-Jade Butkow, Cecilia Mascolo, **Yang Liu**. "IMChew: Chewing Analysis using Earphone Inertial Measurement Units", BodySys, 2024.

[14] Kayla-Jade Butkow, Ting Dang, Andrea Ferlini, Dong Ma, **Yang Liu**, Cecilia Mascolo. "An Evaluation of Heart Rate Monitoring with In-ear Microphones under Motion", Pervasive and Mobile Computing Journal, 2024. [Media Exposure: "SMU City Perspectives"]

[15] Jiao Li*, Jiakai Xu, **Yang Liu**, Weitao Xu, Zhenjiang Li. "Enhancing the Applicability of Sign Language Translation", IEEE Transactions on Mobile Computing, 2024.

[16] Jacob Brown*, **Yang Liu**, Cecilia Mascolo. "Yawning Detection using Earphone Inertial Measurement Units", ACM SmartWear, 2023.

[17] Jiani Cao*, Chengdong Lin, **Yang Liu**, Kun Wang, Zhenjiang Li. "Poster: Gaze Tracking on Any Surface with Your Phone", ACM TURC, 2023.

[18] Jiao Li*, **Yang Liu**, Zhenjiang Li, Jin Zhang. "EarPass: Continuous User Authentication with In-ear PPG", EarComp, 2023. [Best Paper Award]

[19] Jiani Cao*, Chengdong Lin, **Yang Liu**, Zhenjiang Li. "Gaze Tracking on Any Surface with Your Phone", ACM SenSys, 2022.

[20] Jiao Li*, **Yang Liu**, Weitao Xu, Zhenjiang Li. "GASLA: Enhancing the Applicability of Sign Language Translation", IEEE INFOCOM, 2022.

[21] Zhen Xiao, Tao Chen, **Yang Liu**, Zhenjiang Li. "Keystroke Recognition with the Tapping Sound Recorded by Mobile Phone Microphones", IEEE Transactions on Mobile Computing, 2021.

[22] **Yang Liu**, Chengdong Lin, Zhenjiang Li. "WR-Hand: Wearable Armband can Track User's Hand", ACM IMWUT journal (UbiComp) 2021.

[23] Jiao Li, **Yang Liu**, Tao Chen, Zhen Xiao, Zhenjiang Li, Jianping Wang. "Adversarial Attacks and Defenses on Cyber-Physical Systems: A Survey", IEEE Internet of Things Journal, 2020.

[24] Zhen Xiao, Tao Chen, **Yang Liu**, Zhenjiang Li. "Mobile Phones Know Your Keystrokes through the Sounds from Finger's Tapping on the Screen", IEEE ICDCS, 2020.

[25] **Yang Liu**, Zhenjiang Li. "aLeak: Context-Free Side-Channel from Your Smart Watch Leaks Your Typing Privacy", IEEE Transactions on Mobile Computing, 2019.

[26] **Yang Liu**, Zhenjiang Li, Zhidan Liu, Kaishun Wu. "Real-time Arm Skeleton Tracking and Gesture Inference Tolerant to Missing Wearable Sensors", ACM MobiSys, 2019.

[27] **Yang Liu**, Chengdong Lin, Zhenjiang Li, Zhidan Liu, Kaishun Wu. "Poster: When Wearable Sensing Meets Arm Tracking", ACM MobiSys WKSHPS, 2019.

[28] **Yang Liu**, Yonghang Jiang, Zhenjiang Li, Jianping Wang. "Rulers on Our Arms: Waving to Measure Object Size through Contactless Sensing", ACM Transactions on Sensor Networks, 2019.

[29] Kang Yang, Tianzhang Xing, **Yang Liu**, Zhenjiang Li, Xiaoqing Gong, Xiaojiang Chen, Dingyi Fang. "cDeepArch: A Compact Deep Neural Network Architecture for Mobile Sensing", IEEE/ACM Transactions on Networking, 2019.

[30] Yonghang Jiang, **Yang Liu**, Zhenjiang Li. "Data Fusion and Alignment for Location-Aware Crowdsourcing Applications", IEEE DySPAN, 2019.

[31] **Yang Liu**, Zhenjiang Li. "aLeak: Privacy Leakage through Context-Free Wearable Side-Channel", IEEE INFOCOM, 2018. [Best-in-Session Presentation Award]

[32] **Yang Liu**, Zhenjiang Li. "Poster Abstract: Context-Free Wearable Side-Channel Leaks Your Typing Privacy", IEEE INFOCOM WKSHPS, 2018.

[33] Kang Yang, Xiaoqing Gong, **Yang Liu**, Zhenjiang Li, Tianzhang Xing, Xiaojiang Chen, Dingyi Fang. "cDeepArch: A Compact Deep Neural Network Architecture for Mobile Sensing", IEEE SECON, 2018.